

WASP-140b

R-2046

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_250111 · created 2026-08-08T23:40:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460686.6938 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.022
Transit depth (Rp/Rs)^2	2.92 +/- 0.75 [%]
Orbital Inclination (inc)	83.42 +/- 0.77
Airmass coefficient 1 (a1)	1.228 +/- 0.012
Airmass coefficient 2 (a2)	-0.117 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	None
Optimal Aperture	4.59
Optimal Annulus	9.14
Transit Duration (day)	0.065 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.022 vs 0.1432 (deviation 1.26 σ)
Duration fit vs published	0.065 d vs 0.0604 d (deviation 0.31 σ)
Mid-time O–C	0.3 min
Depth SNR	7.8
Residual scatter	0.98 %

Light curve

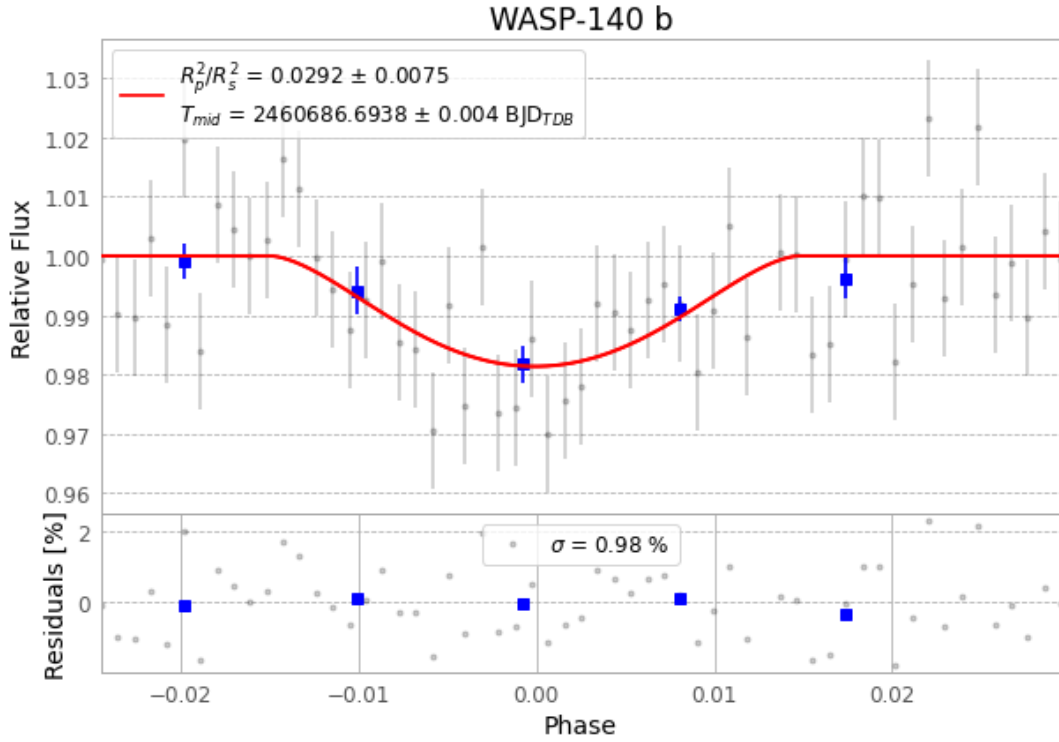


Figure 1 — FinalLightCurve_WASP-140 b_2025-01-10.png (SHA-256 f1eafb61bfb5e740...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 be7c57ba6270e99b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

59 FITS frames (39 MB), WASP-140250111031515.FITS → WASP-140250111060915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-250111.log (1d1d973cbb18...), FinalLightCurve_WASP-140 b_2025-01-10.png (f1eafb61bfb5...), FinalLightCurve_WASP-140 b_2025-01-10.csv (2c90002d9cf1...)

Compute resources

Wall time 294.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:40Z.